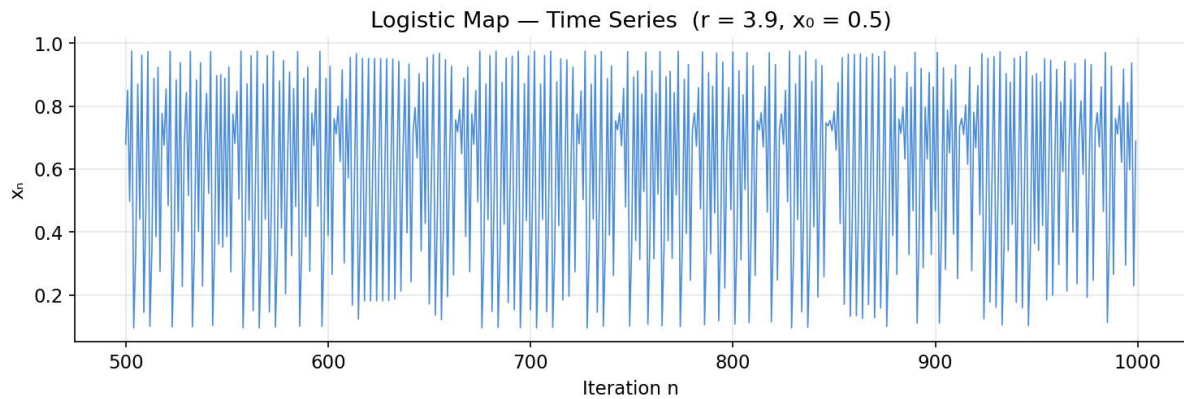


Name: Rana Mohamed Ahmed Shaqr

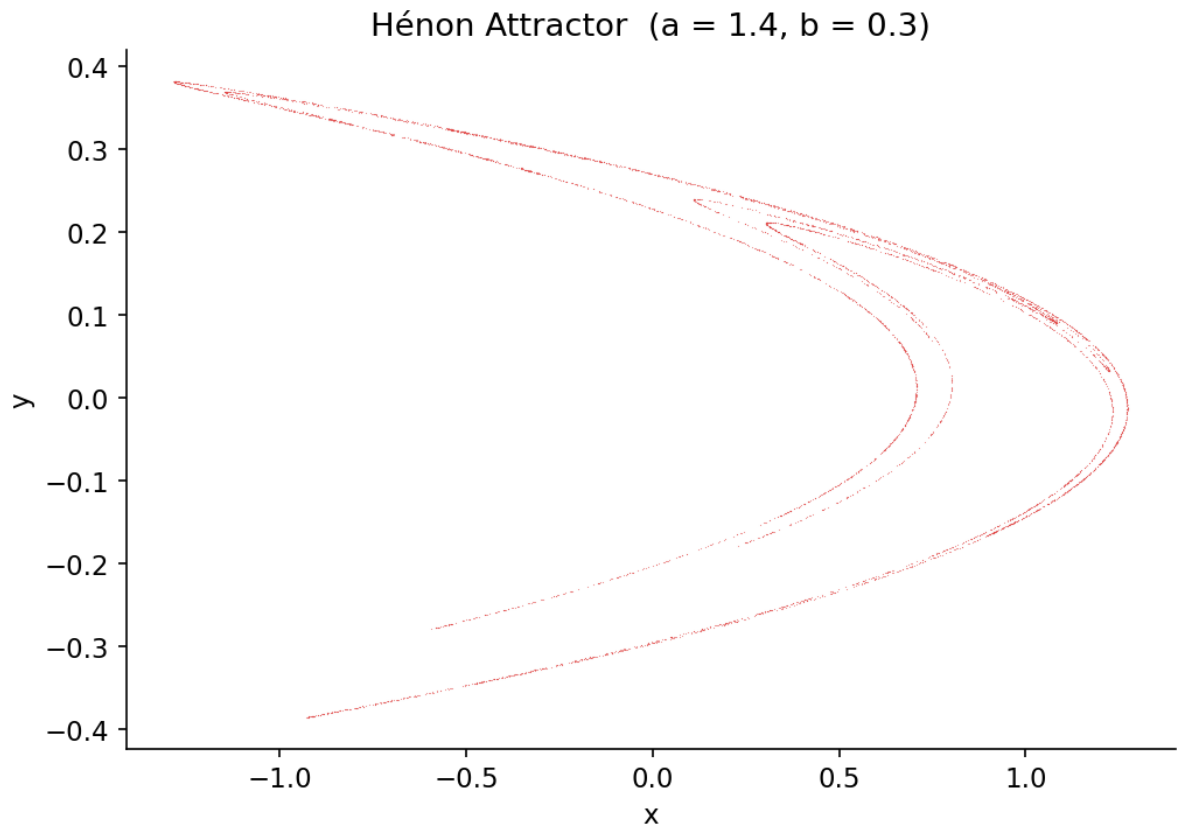
Group: LLM

Task 1

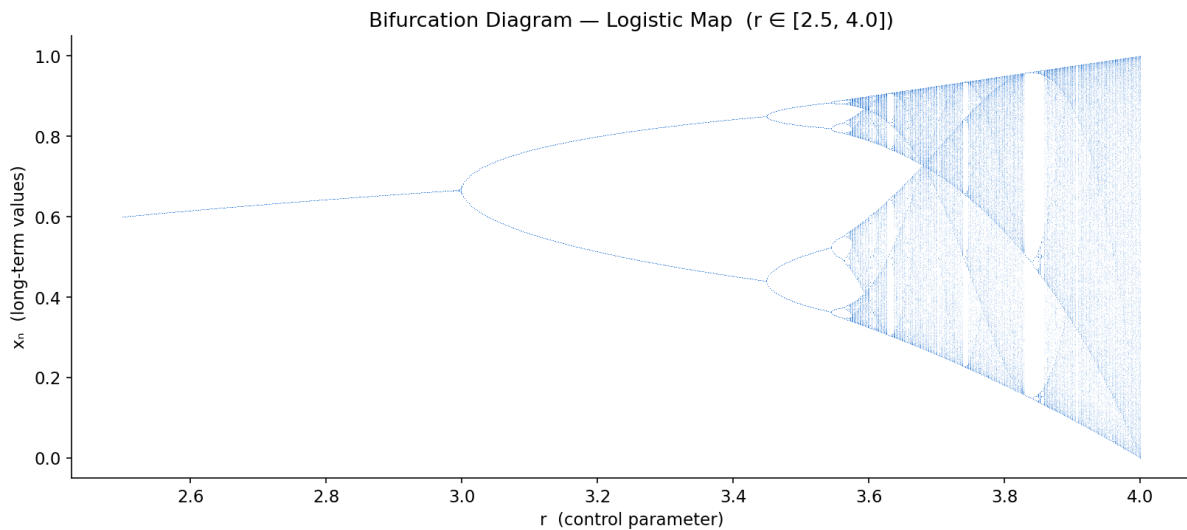
Part A:



x_n oscillates between 0 and 1 (min, max) non-periodically and doesn't reach steady state, so it means it still deterministic but sensitive to initial conditions.



The scatter of iterations is thin curved line which is fractal attractor as it's not scattered all over plane they all gathered in these curved lines so it's fractal between 1 and 2.

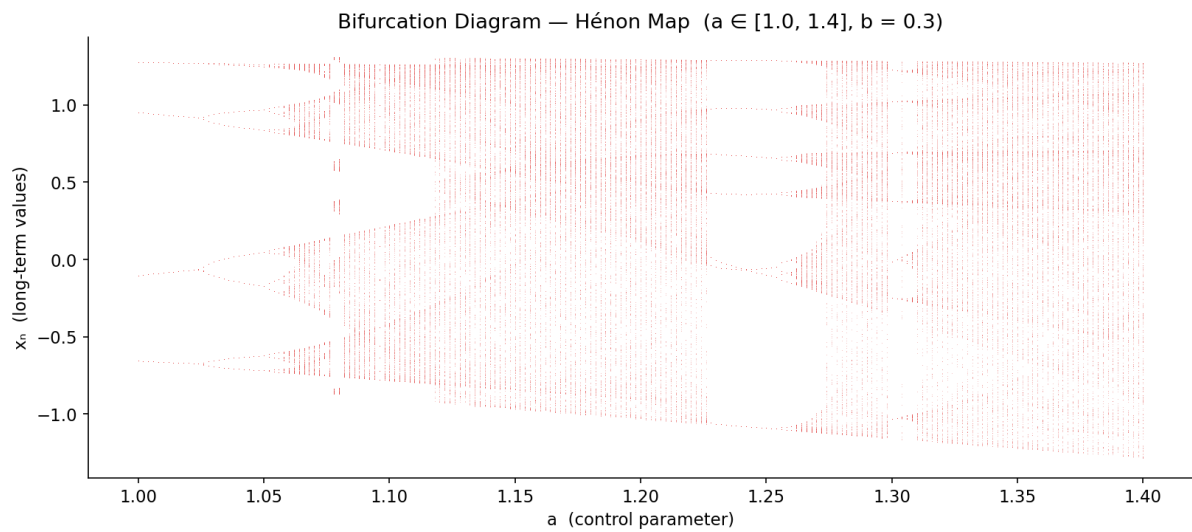


$r < 3$: system is stable

$3 < r < 3.45$ system oscillates with period 2

Then becomes 4 periods then 8 in a period-doubling cascade

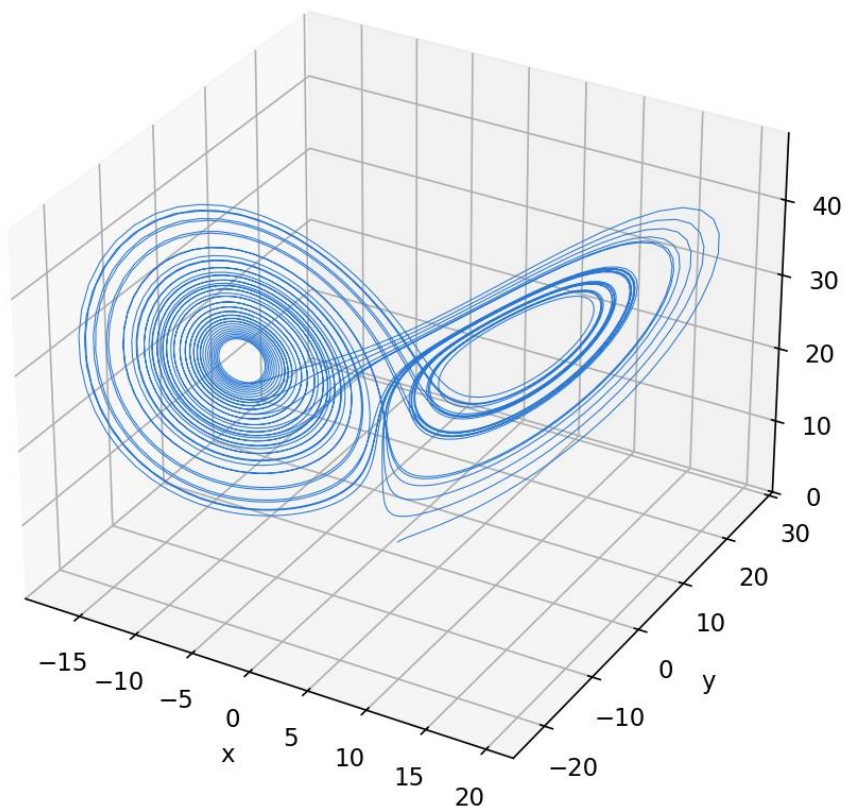
Then branches merge into dense chaotic bands but there are white gaps where system become periodic (predictable)

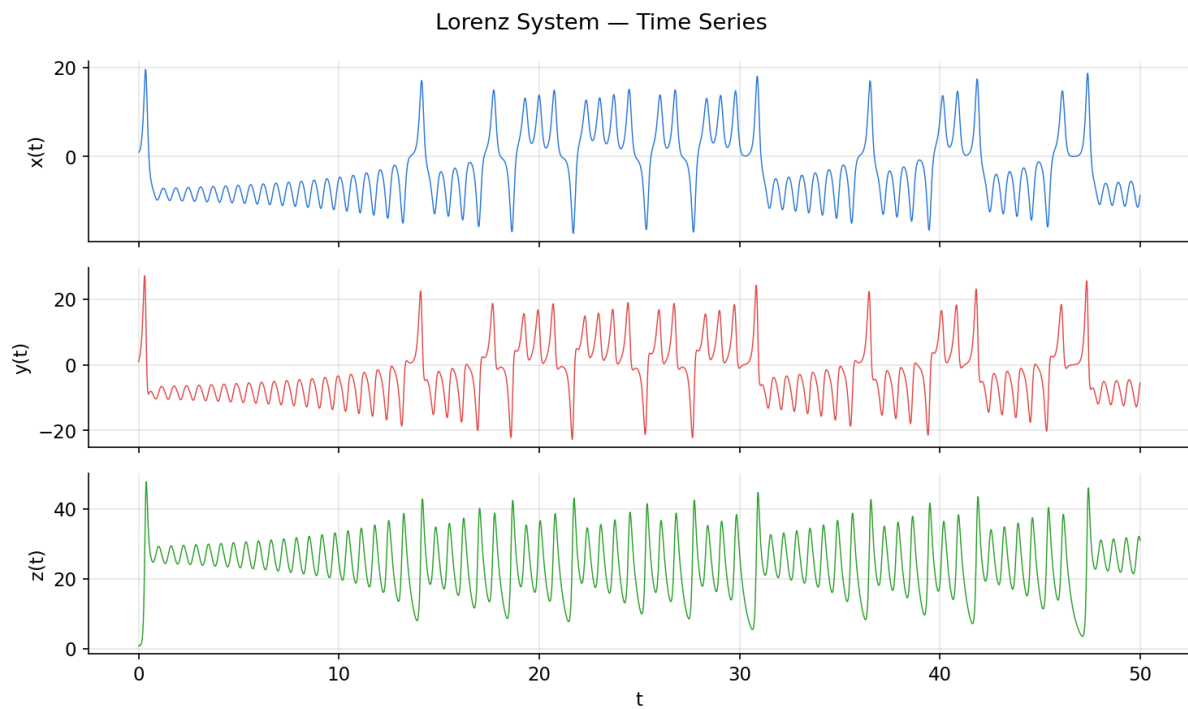


For small a a system is periodic then continue period-doubling cascade until branches merge into broad chaotic bands and there are also gaps when it became periodically again

Part B:

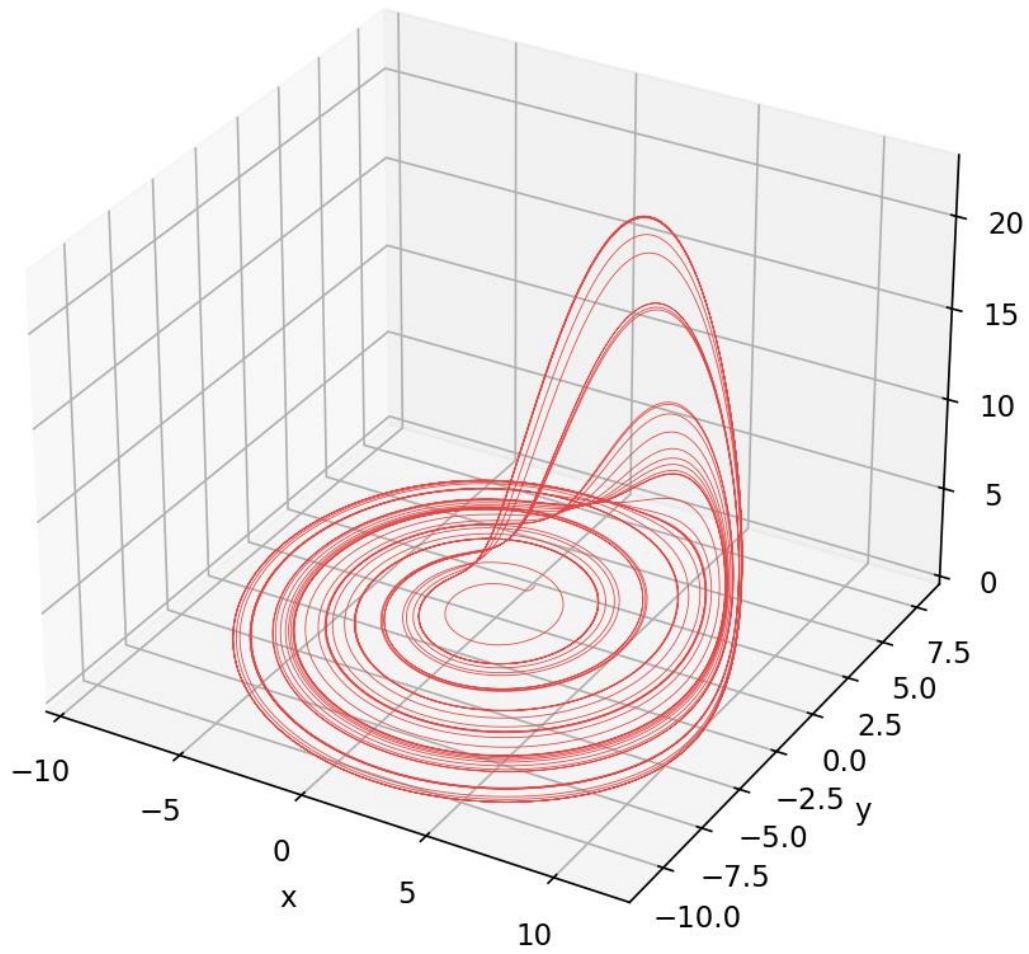
Lorenz Attractor ($\sigma=10$, $\rho=28$, $\beta=8/3$)

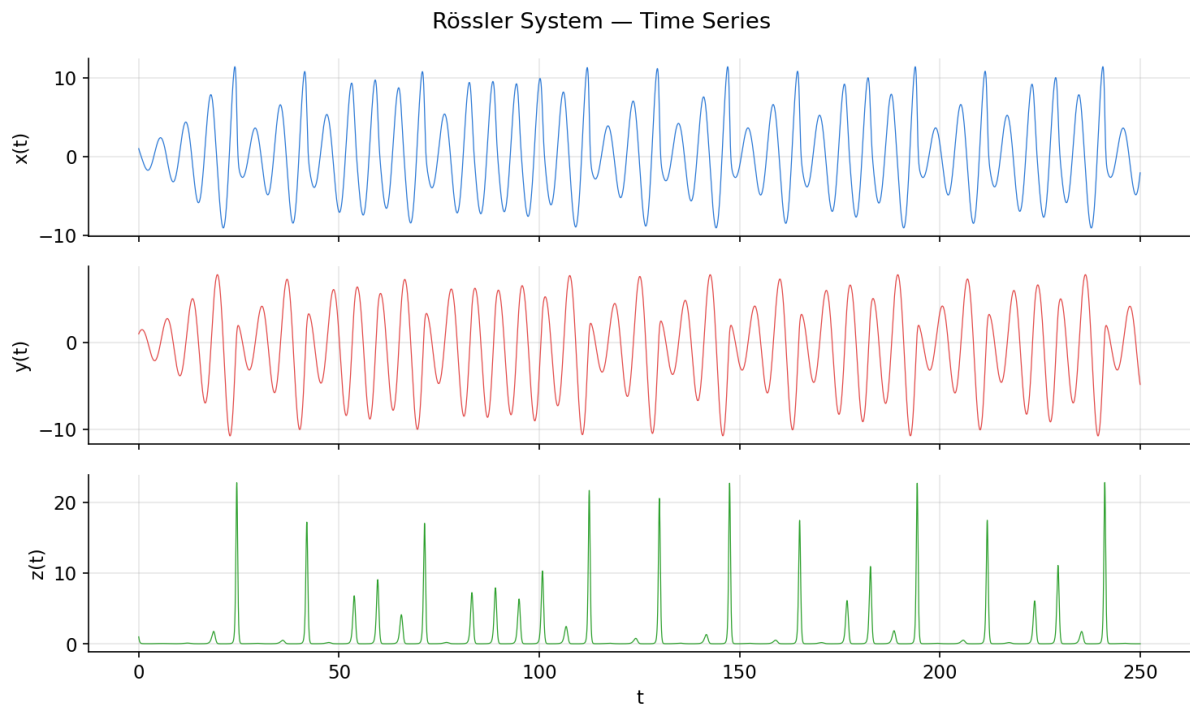




butterfly attractor: the trajectory spirals tightly around one wing for a while, then unpredictably jumps across to spiral around the other wing and not repeating same path. We can see in time series that x and y that it flips from $-ve$ to $+ve$ showing going from first wing to second wing when radius of spiral is small at beginning.

Rössler Attractor ($a=0.2$, $b=0.2$, $c=5.7$)

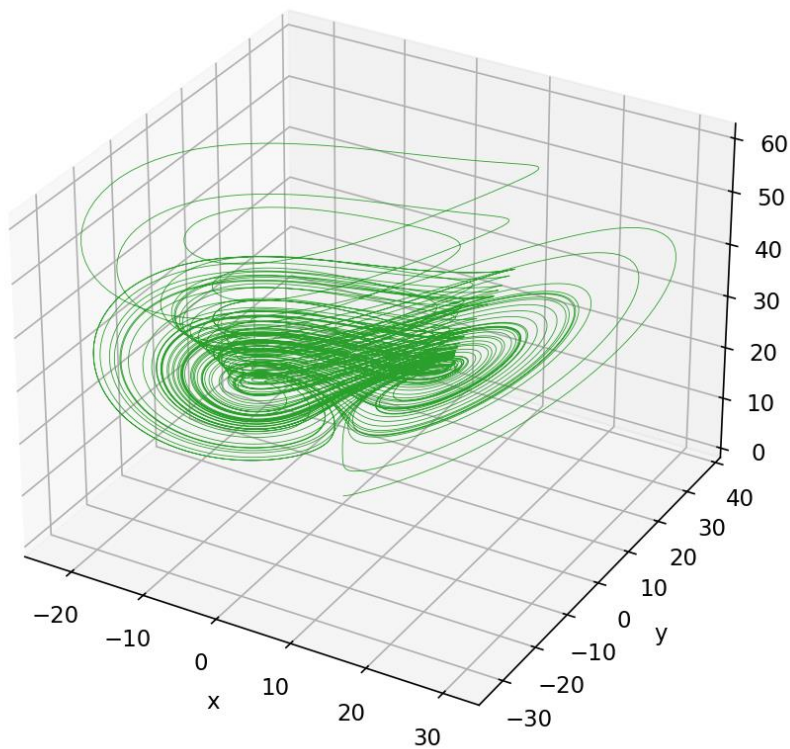


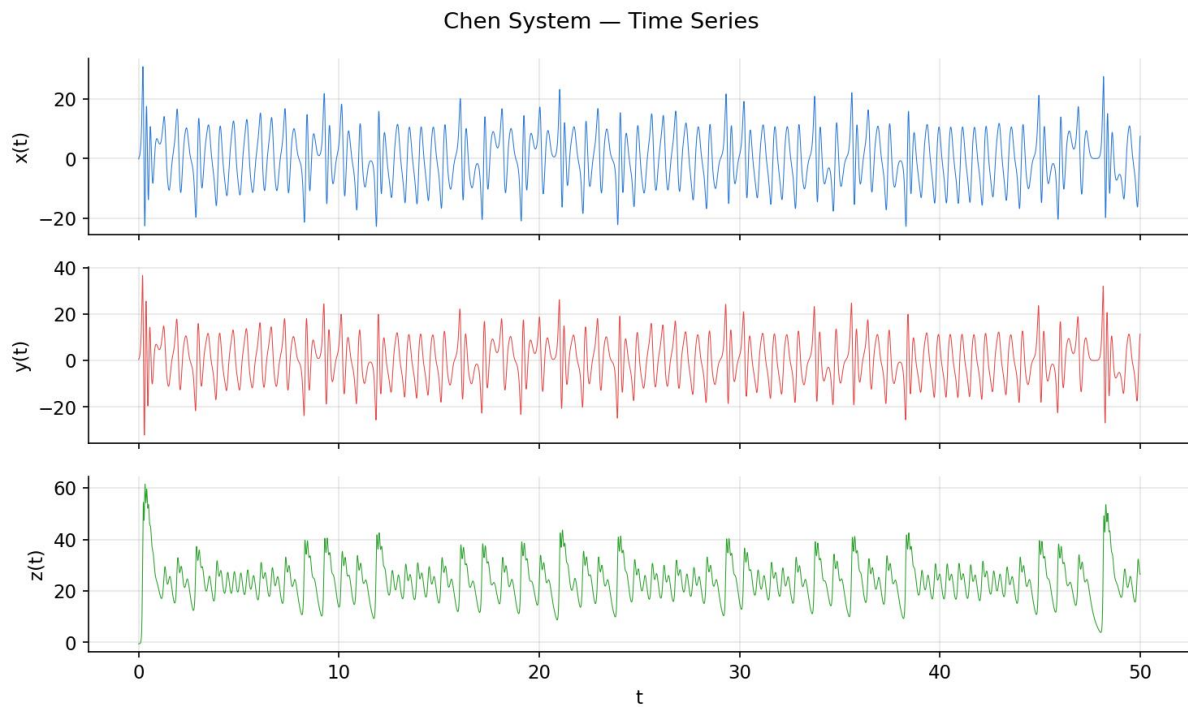


single-scroll attractor

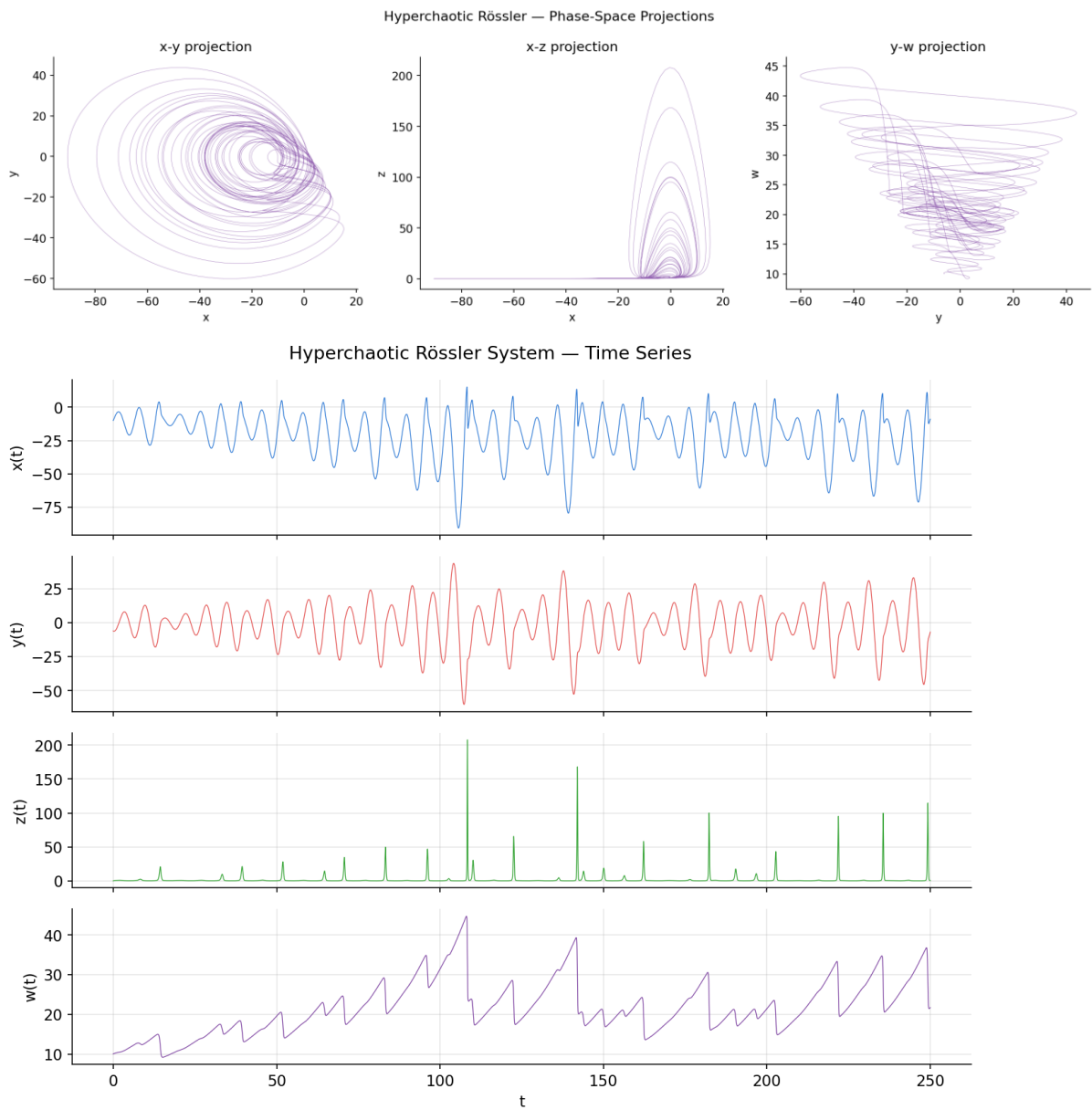
Chaos is visible at z which is near zero but spikes irregularly.

Chen Attractor ($a=35$, $b=3$, $c=28$)





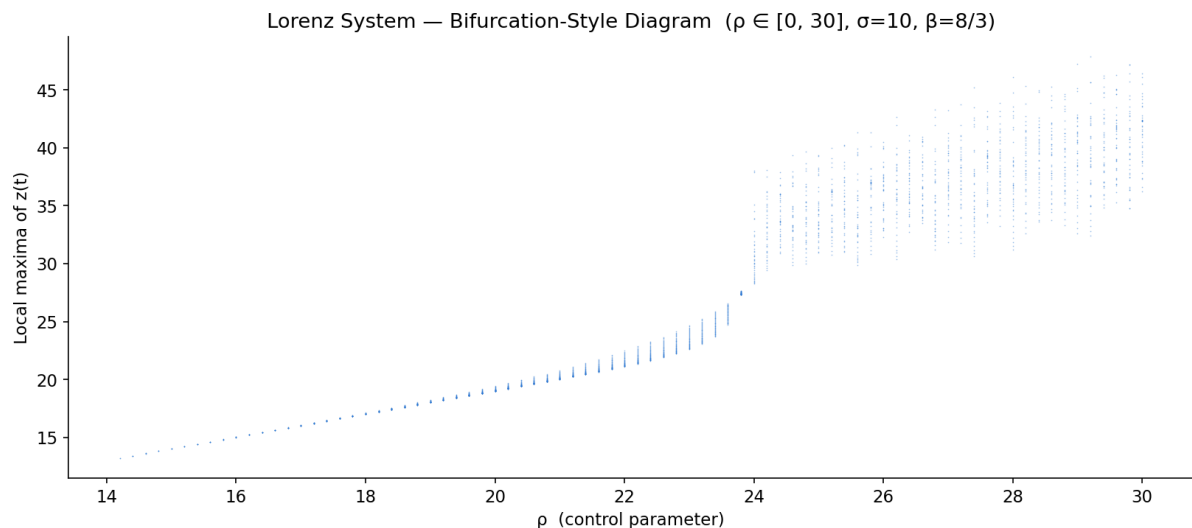
Double-scroll attractor but denser weaving due to stiffer coefficients
sign flips in $x(t)$ and $y(t)$ show change in spiral and the transients are sharp and the spikes large. $z(t)$ is around 20 but spike is larger. This shows that system is faster and more chaotic.



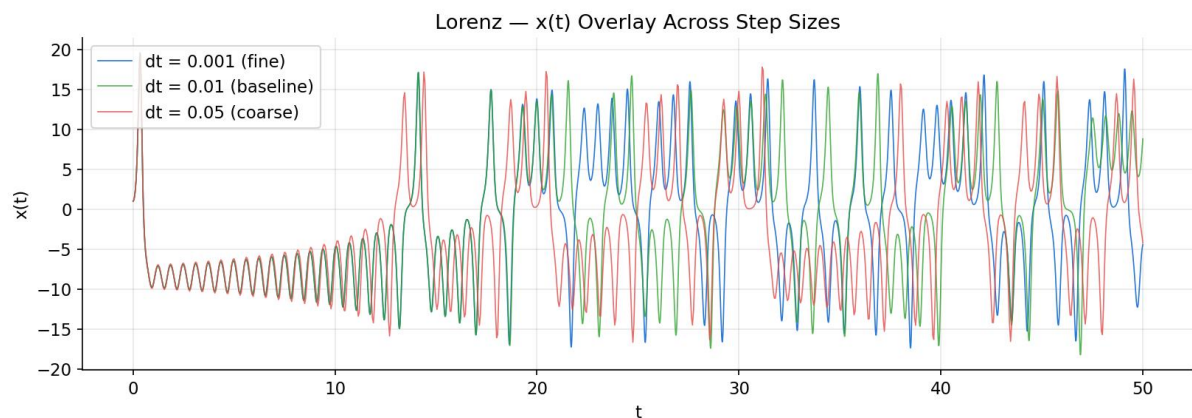
x, y, z act like Rossler system.

w is a repeating sawtooth and it drops sharply every time the z spikes due to equation as small $+d \cdot w$ term slowly drives w up whenever z is near zero, and the $-c \cdot z$ term brings it back down whenever z spikes.

Part C:

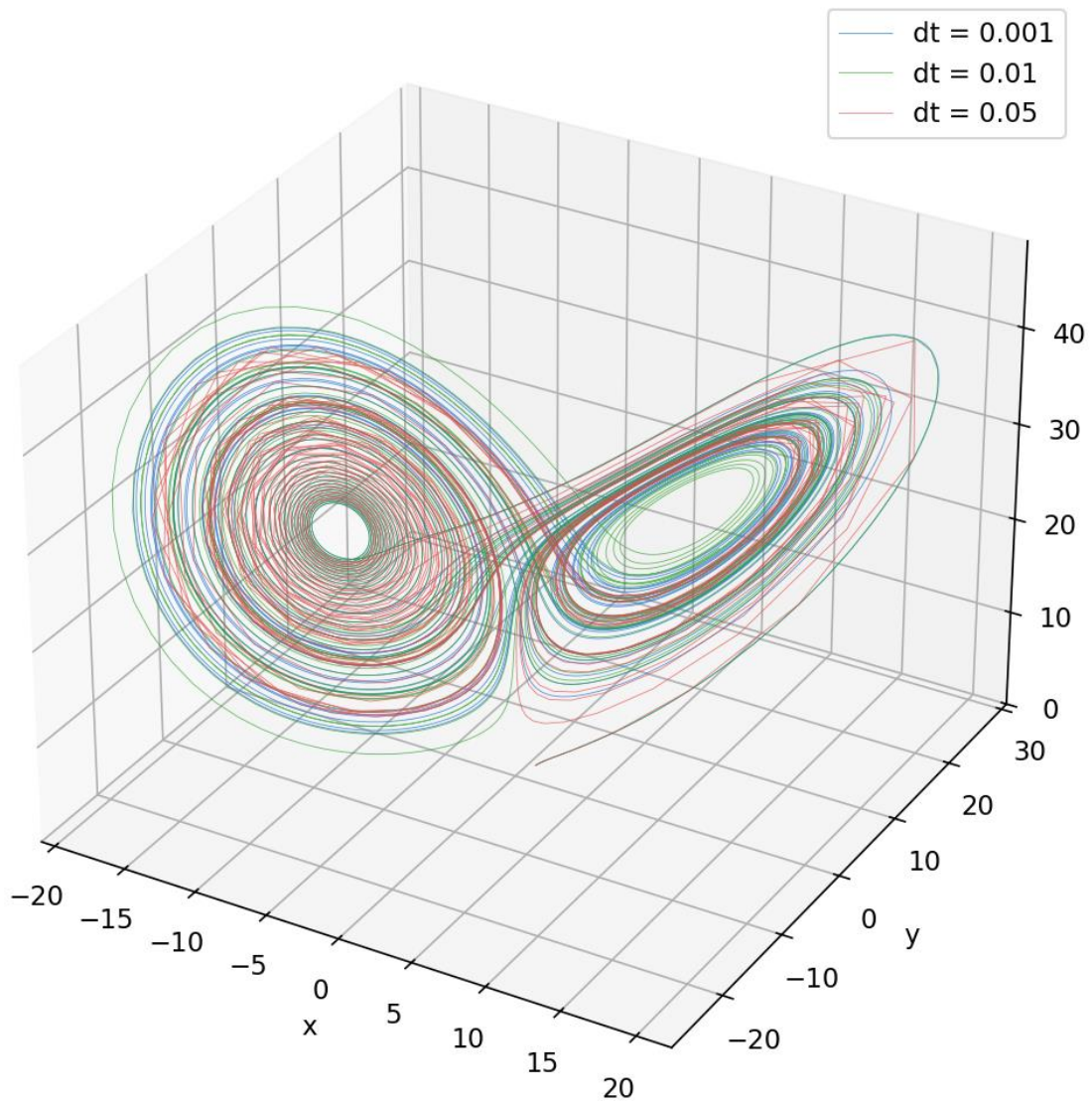


Below $p=14$, there is no real peak so this means there was no real peak, so it was predictable. At $14 < p < 24$, rising lines and there are few peaks, but system is still predictable. At $P > 24$, sharp transition happens and there are many peaks for each value of p meaning that system entered chaos state.

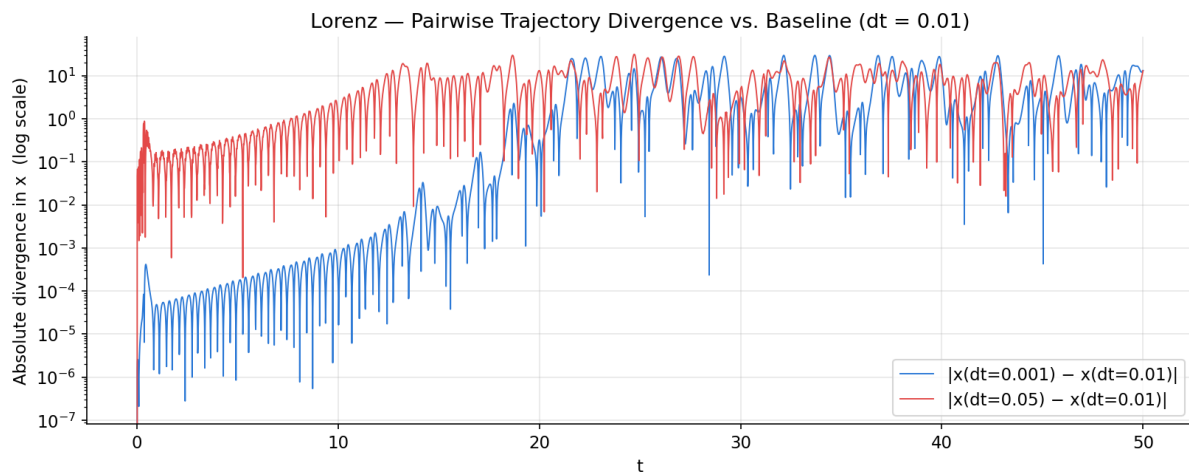


At start all three lines are almost identical then a shift between base and coarse happening around $t = 12$, around $t=22$ all three plots become decorrelated although all three started from the identical initial condition and that's because no matter how small dt is, it introduces a tiny approximation error (RK4 fits a polynomial to the true curve over each step, and a coarser dt fits it more loosely)

Lorenz — 3-D Trajectory Overlay Across Step Sizes



All three step sizes follow same butterfly shape, so attractor is still same even for large time steps which means it is insensitive feature, but the only difference is the trajectory (values of system at each timestep) as tiny numerical error introduced by a larger dt is amplified by the system's chaotic sensitivity causing the trajectories to diverge from one another even while remaining on the same attractor.



dt=0.05 introduces much more per-step error than dt=0.001 as it introduces much more per-step error from start

dt=0.01 is trustworthy against dt=0.001 only up to about t=15 after which the accumulated divergence grows largely.

dt=0.05 is unreliable

Discrete vs. continuous chaos:

Both reach the same chaotic state due to sensitive dependence on initial conditions but arrive there differently. Due to converting continuous system to discrete system, it adds errors to system unlike discrete systems which isolate chaos as a pure property of the map's rule as true system is similar to simulation. Continuous chaotic systems are chaotic in two compounding senses: dynamically (with respect to initial conditions) and numerically (with respect to integration accuracy).

Chaotic vs. hyperchaotic:

Both has same conditions that one or more positive Lyapunov exponents causing nearby trajectories to stretch apart, but differ in how many independent directions that stretching happens in. Ordinary chaos needs only one direction, which is why a 3-D phase space (3 state variables) is enough to contain it. Hyperchaos requires a second, independent stretching direction, this is why hyperchaotic systems need at least a 4th state variable. Hyperchaotic systems are therefore unpredictable in two compounding senses: along the first unstable direction (as in ordinary chaos), and along a second, independent unstable direction that ordinary chaotic systems don't have.